26F4 Homework

Integration of Knowledge and Ideas and Geometric Figures and Angles

What will homework look like?

Every week, we will give you 6–8 questions to work on (about 30 minutes), split between English language arts and mathematics. Some questions will be more challenging than others.

Try your best to submit your work before the start of the next lesson, but remember that homework is optional. You will sometimes see additional practice worksheets. You do not need to complete those; we will only grade the documents labeled "26F*X* Homework."

Homework will be graded out of eight points; a tutor will give you feedback on your work. Feel free to resubmit and correct your mistakes as many times as needed.

English Language Arts: Key Ideas and Details

The Case for Later School Start Times

1 Many middle and high schools begin classes early in the morning, requiring students to wake up before sunrise. Some educators argue that delaying school start times would improve students' academic performance and overall well-being. However, others question whether changing school schedules would produce meaningful educational benefits.

2 Supporters of later start times point to the relationship between sleep and learning. Sleep plays an important role in memory consolidation, the process through which the brain strengthens and organizes information acquired

during the day. When students consistently receive insufficient sleep, they may struggle to maintain attention, remember information, and complete complex tasks. Consequently, advocates argue that allowing students to sleep longer could improve their readiness to learn.

3 In one school district, administrators observed that the percentage of students arriving late decreased after the district delayed the beginning of its school day by 45 minutes. Teachers also reported that students appeared more attentive during morning lessons. Although these observations suggest potential benefits, the district did not compare academic achievement before and after the schedule change. Therefore, the findings do not establish whether starting school later improved students' academic performance.

4 Opponents argue that later start times could create logistical difficulties. School districts might need to adjust bus schedules, while students participating in athletics or after-school activities could face scheduling conflicts. Additionally, families relying on older students to care for younger siblings might struggle to accommodate changes to dismissal times.

5 Despite these concerns, schools should consider experimenting with later start times. However, before implementing permanent changes, administrators should examine evidence about academic performance, student health, and scheduling challenges. A carefully designed trial could help determine whether the benefits justify the disruptions.

1. How does the author develop the argument that schools should consider later start times? **[1]**

A. By examining the relationship between sleep and learning, presenting observations from one district, acknowledging logistical concerns, and recommending further evaluation.

B. By discussing the history of school schedules and demonstrating that traditional start times are responsible for declining academic performance.

C. By comparing academic achievement across several school districts and concluding that later start times consistently improve examination scores.

D. By identifying objections to later start times and demonstrating that these concerns have no reasonable justification.

Justify your answer here.

2. Which detail from the passage provides the strongest direct evidence that changing school start times may improve students' punctuality? **[1]**

A. Sleep plays an important role in strengthening and organizing information.

B. Some students participate in athletic programs and other after-school activities.

C. A school district reported a decrease in student tardiness after delaying its start time by 45 minutes.

D. Teachers observed that students appeared more attentive during morning classes.

Justify your answer here.

3. Which statement best evaluates the evidence presented in paragraph 3? **[1]**

A. The evidence is sufficient to demonstrate that later start times significantly improve academic achievement because teachers reported greater attentiveness.

B. The evidence is irrelevant because observations about attendance and attentiveness have no relationship to students' educational experiences.

C. The evidence establishes that later start times benefit every student because fewer students arrived late after the schedule change.

D. The evidence suggests possible improvements in punctuality and attentiveness but is insufficient to establish improvements in academic achievement.

Justify your answer here.

4. Which additional evidence would most strongly support the author's recommendation that schools experiment with later start times?

A. A survey indicating that most students prefer waking up later in the morning.

B. A controlled study finding that students at schools adopting later start times showed greater improvements in academic performance and

sleep duration than comparable students at schools maintaining earlier schedules.

C. Interviews with parents who believe that school districts should prioritize student happiness when making scheduling decisions.

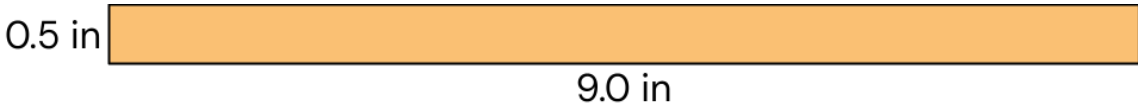
D. A report showing that school buses are one of the largest transportation expenses for many school districts.

Justify your answer here.

Mathematics: Transformations

Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. You may not use a calculator. For all questions in this part, a correct answer with no work shown will receive only **0.5 point**.

5. The drawing below represents a rectangle in real life using the scale factor 1 inch : 1 meter. Identify the area of the real-life rectangle to the nearest tenth of a meter. **[1]**



Write your answer here.

6. Which of the following triangles is possible to draw? (There may be multiple correct answers.) **[1]**

A. A triangle with side lengths 14, 19, and 4

B. A triangle with side lengths 8, 6, and 12

C. An acute triangle with angle measures $3°$ and $32°$

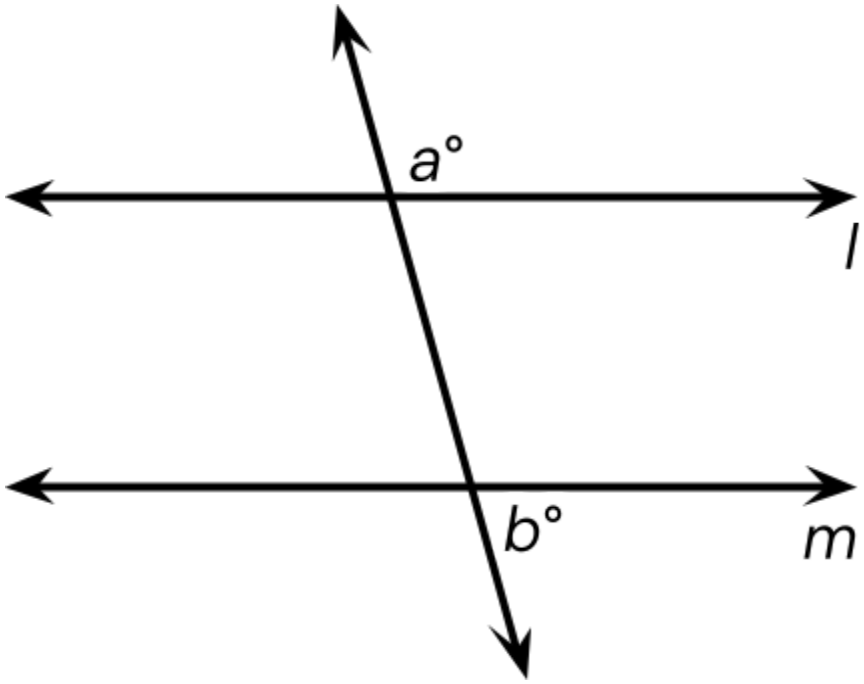
D. A right triangle with an angle measure $86°$

Justify your answer here.

7. A right square pyramid with a base area of 25 meters^2 and a height of 3 meters is sliced parallel to the base, 2 meters above the base. What is the perimeter of the cross section? **[1]**

Write your answer here.

8. In the figure below, $l \parallel m$. If $a = 4x$ and $b = 3x + 54$, solve for a.



Write your answer here.